Fixture record fx-0002

Title: System dynamics simulation of surge capacity in rural disaster response

Abstract:
We build a system-dynamics model of hospital surge capacity in three rural US counties during simulated hurricane evacuation scenarios, testing staffing and bed-allocation policies under demand shocks.

Methods
Study design: Simulation study (system dynamics model)
Country: United States
Population: Three rural county hospital systems and their catchment populations
Concept: Surge-capacity system dynamics simulation for hospital bed and staffing allocation
Context: Rural hurricane-evacuation disaster response

Results
- Dynamic bed-reallocation policies reduced simulated patient wait times by 18% compared to static allocation.
- Staffing shortfalls emerged earliest in the smallest of the three simulated county hospitals.

Limitations
The authors state the model was calibrated against a single historical hurricane season and may not generalise to slower-onset disasters.

Funding
Funding sources: National Institute for Rural Health Systems (fictional, synthetic fixture)
Funder role: The funder participated in study design by specifying which hurricane scenarios to simulate, but had no role in data analysis or the decision to publish.

Synthetic test data — pipeline conformance fixture, not scientific evidence.